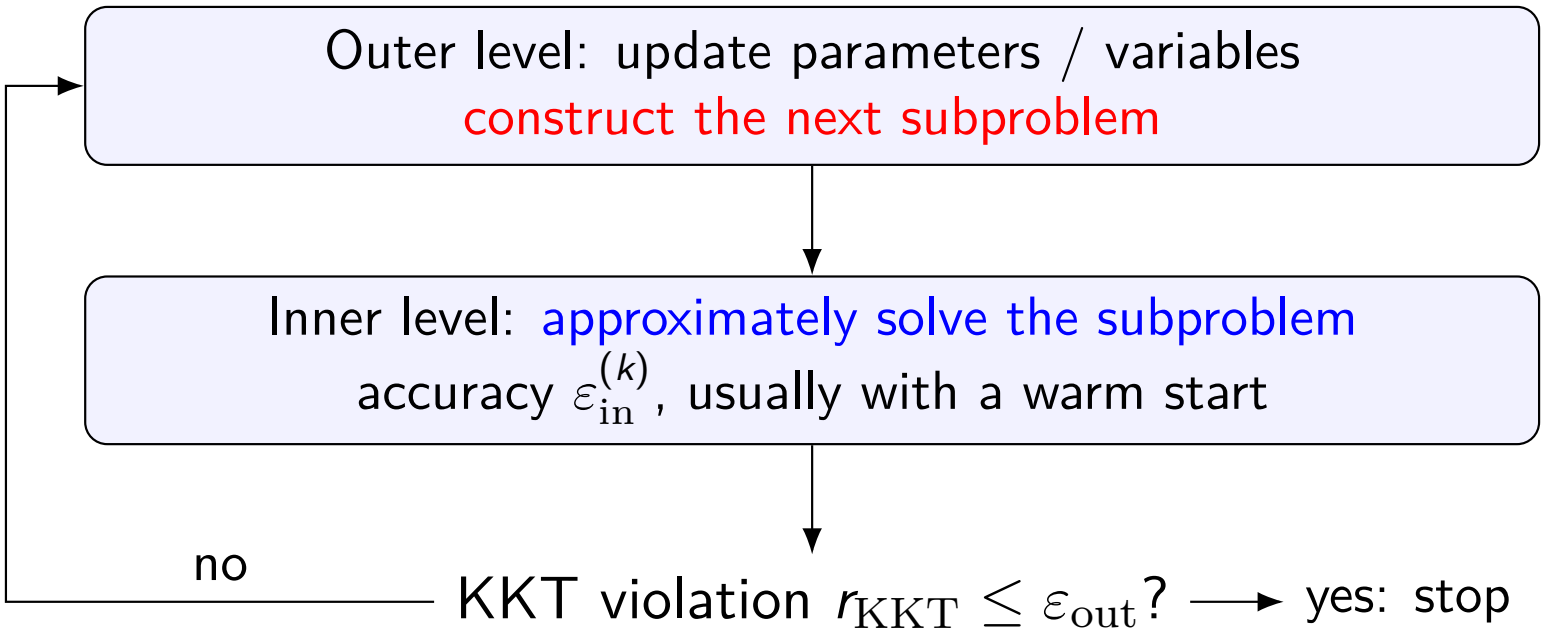
Lecture 16: Lagrange–Newton and Sequential Quadratic Programming

MA3236 Non-Linear Programming

Hu Yukuan

yukuan.hu@nus.edu.sg
S17-08-21

Quick review: the two-level framework



- The usual target is an **approximate KKT point**.
- Balance outer accuracy and inner error.

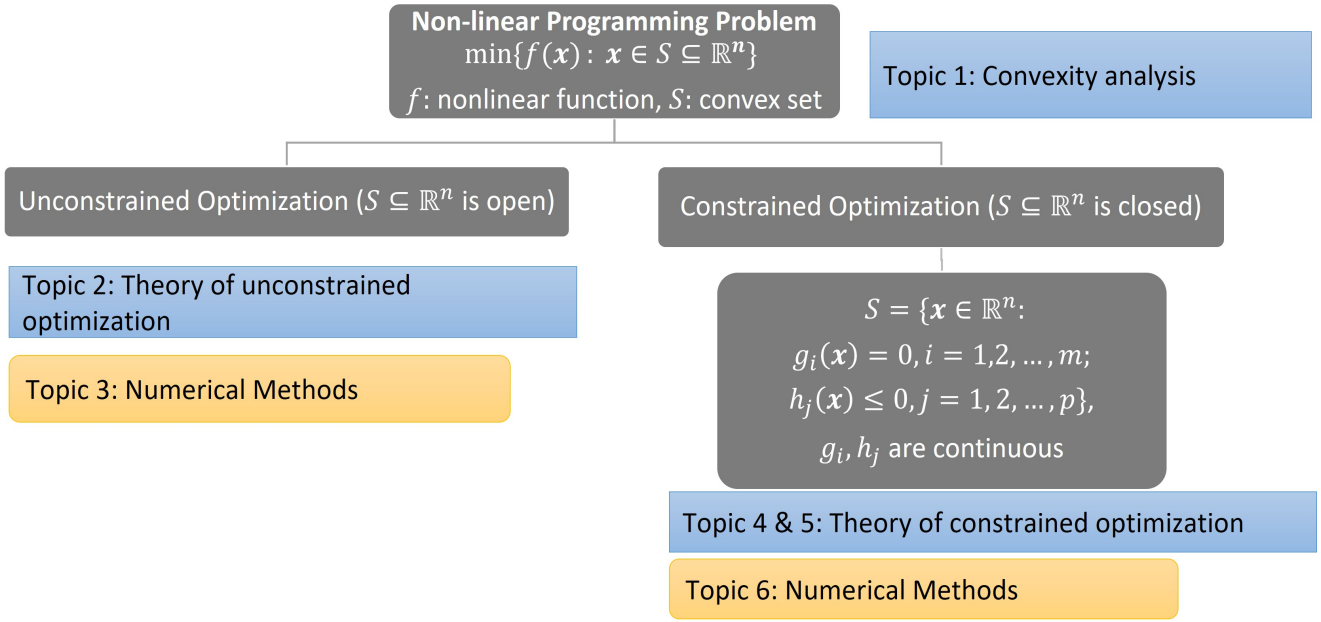
Quick review: QP and ALM for equality constraints

For $\mathbf{g}(\mathbf{x}) = \mathbf{0}$, both methods solve **unconstrained** subproblems.

	subproblem objective	multiplier estimate
QP	$f(\mathbf{x}) + \frac{\rho^{(k)}}{2}\|\mathbf{g}(\mathbf{x})\|^2$	$\boldsymbol{\lambda}^{(k+1)} = \rho^{(k)}\mathbf{g}(\mathbf{x}^{(k+1)})$
ALM	$f(\mathbf{x}) + \boldsymbol{\lambda}^{(k)T}\mathbf{g}(\mathbf{x}) + \frac{\rho}{2}\|\mathbf{g}(\mathbf{x})\|^2$	$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho\mathbf{g}(\mathbf{x}^{(k+1)})$

- QP: typically $\rho^{(k)} \uparrow \infty$ to make feasibility violation small.
- ALM: under suitable local regularity, a fixed, sufficiently large ρ can work.
- Balance outer accuracy, inner error, and **penalty parameter updates**.
- This lecture: retain linearized constraints in the subproblem.

Topics



In Topic 6, we study numerical methods for solving **constrained optimization problems**.

- Quadratic penalty and augmented Lagrangian methods
- Lagrange–Newton and sequential quadratic programming
- Barrier methods
- Subgradient methods for the dual problem

Equality KKT conditions as algebraic equations

$$\min_{\mathbf{x}\in\mathcal{D}} f(\mathbf{x}) \quad \text{s. t.} \quad \mathbf{g}(\mathbf{x}) = \mathbf{0}, \qquad \mathcal{D} \subseteq \mathbf{R}^n \text{ open.}$$

Write

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}), \qquad \mathbf{A}(\mathbf{x}) = [\nabla g_1(\mathbf{x}), \ldots, \nabla g_m(\mathbf{x})].$$

$$\boxed{\begin{pmatrix} \nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}) \\ \mathbf{g}(\mathbf{x}) \end{pmatrix} = \mathbf{0}.}$$

$$r_{\text{stat}} = \|\nabla_{\mathbf{x}} L\|_\infty, \quad r_{\text{feas}} = \|\mathbf{g}\|_\infty, \quad r_{\text{KKT}} = \max\{r_{\text{stat}}, r_{\text{feas}}\}.$$

A KKT pair

Equality KKT conditions as algebraic equations

$$\min_{\mathbf{x}\in\mathcal{D}} f(\mathbf{x}) \quad \text{s. t.} \quad \mathbf{g}(\mathbf{x}) = \mathbf{0}, \qquad \mathcal{D} \subseteq \mathbf{R}^n \text{ open.}$$

Write

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}), \qquad \mathbf{A}(\mathbf{x}) = [\nabla g_1(\mathbf{x}), \ldots, \nabla g_m(\mathbf{x})].$$

With $\mathbf{z} = (\mathbf{x}, \boldsymbol{\lambda})$, define

$$\boxed{\mathbf{F}(\mathbf{z}) = \begin{pmatrix} \nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}) \\ \mathbf{g}(\mathbf{x}) \end{pmatrix} = \mathbf{0}.}$$

$$r_{\text{stat}} = \|\nabla_{\mathbf{x}} L\|_\infty, \quad r_{\text{feas}} = \|\mathbf{g}\|_\infty, \quad r_{\text{KKT}} = \max\{r_{\text{stat}}, r_{\text{feas}}\}.$$

A KKT pair $\Leftrightarrow$ a root of $\mathbf{F}$ $\Rightarrow$ apply Newton's method.

When Lagrange meets Newton

For a correction $\mathbf{s} = (\mathbf{d}, \delta)$,

$$\mathbf{F}(\mathbf{z}) = \begin{pmatrix} \nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}) \\ \mathbf{g}(\mathbf{x}) \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad \mathbf{F}(\mathbf{z} + \mathbf{s}) \approx \mathbf{F}(\mathbf{z}) + \mathbf{J}_F(\mathbf{z})\mathbf{s}.$$

Let $\mathbf{H}_L = \nabla^2_{\mathbf{xx}} L$. Then

$$\mathbf{H}_L = \mathbf{H}_f + \sum_{i=1}^{m} \lambda_i \mathbf{H}_{g_i}, \qquad \mathbf{J}_F(\mathbf{z}) = \begin{pmatrix} \mathbf{H}_L & \mathbf{A} \\ \mathbf{A}^T & \mathbf{0} \end{pmatrix}.$$

When Lagrange meets Newton

For a correction $\mathbf{s} = (\mathbf{d}, \delta)$,

$$\mathbf{F}(\mathbf{z}) = \begin{pmatrix} \nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}) \\ \mathbf{g}(\mathbf{x}) \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad \mathbf{F}(\mathbf{z} + \mathbf{s}) \approx \mathbf{F}(\mathbf{z}) + \mathbf{J}_F(\mathbf{z})\mathbf{s}.$$

Let $\mathbf{H}_L = \nabla^2_{\mathbf{xx}} L$. Then

$$\mathbf{H}_L = \mathbf{H}_f + \sum_{i=1}^{m} \lambda_i \mathbf{H}_{g_i}, \qquad \mathbf{J}_F(\mathbf{z}) = \begin{pmatrix} \mathbf{H}_L & \mathbf{A} \\ \mathbf{A}^T & \mathbf{0} \end{pmatrix}.$$

Setting the linearized model to zero gives

$$\boxed{\begin{pmatrix} \mathbf{H}_L & \mathbf{A} \\ \mathbf{A}^T & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{d} \\ \delta \end{pmatrix} = - \begin{pmatrix} \nabla_{\mathbf{x}} L \\ \mathbf{g} \end{pmatrix}.}$$

Lagrange (KKT) + Newton (linearization) $\Rightarrow$ **Lagrange-Newton (LN) method**.

Pure LN: pseudocode

Choose $(\mathbf{x}^{(0)}, \boldsymbol{\lambda}^{(0)})$, $\varepsilon_{\text{out}} > 0$, and a step limit. Before each solve, return if $r_{\text{KKT}} \le \varepsilon_{\text{out}}$.

1. Form $\mathbf{H}_L^{(k)}$, $\mathbf{A}^{(k)}$, and solve

$$\begin{pmatrix} \mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d}^{(k)} \\ \boldsymbol{\delta}^{(k)} \end{pmatrix} = -\begin{pmatrix} \nabla_{\mathbf{x}} L^{(k)} \\ \mathbf{g}^{(k)} \end{pmatrix}.$$

Pure LN: pseudocode

Choose $(\mathbf{x}^{(0)}, \boldsymbol{\lambda}^{(0)})$, $\varepsilon_{\text{out}} > 0$, and a step limit. Before each solve, return if $r_{\text{KKT}} \le \varepsilon_{\text{out}}$.

1. Form $\mathbf{H}_L^{(k)}$, $\mathbf{A}^{(k)}$, and solve

$$\begin{pmatrix} \mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d}^{(k)} \\ \boldsymbol{\delta}^{(k)} \end{pmatrix} = -\begin{pmatrix} \nabla_{\mathbf{x}} L^{(k)} \\ \mathbf{g}^{(k)} \end{pmatrix}.$$

2. Check $\mathbf{x}^{(k)} + \mathbf{d}^{(k)} \in \mathcal{D}$, then take the full updates

$$\boxed{\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{d}^{(k)}, \qquad \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \boldsymbol{\delta}^{(k)}.}$$

3. Evaluate r_{KKT}. If $r_{\text{KKT}} \le \varepsilon_{\text{out}}$, return the current pair; otherwise, repeat.

Python codes: `lagrange_newton`, `lagrange_newton_step`.

Pure LN: pseudocode

Choose $(\mathbf{x}^{(0)}, \boldsymbol{\lambda}^{(0)})$, $\varepsilon_{\text{out}} > 0$, and a step limit. Before each solve, return if $r_{\text{KKT}} \le \varepsilon_{\text{out}}$.

1. Form $\mathbf{H}_L^{(k)}$, $\mathbf{A}^{(k)}$, and solve

$$\begin{pmatrix} \mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d}^{(k)} \\ \boldsymbol{\delta}^{(k)} \end{pmatrix} = -\begin{pmatrix} \nabla_{\mathbf{x}} L^{(k)} \\ \mathbf{g}^{(k)} \end{pmatrix}.$$

2. Check $\mathbf{x}^{(k)} + \mathbf{d}^{(k)} \in \mathcal{D}$, then take the full updates

$$\boxed{\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{d}^{(k)}, \qquad \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \boldsymbol{\delta}^{(k)}.}$$

3. Evaluate r_{KKT}. If $r_{\text{KKT}} \le \varepsilon_{\text{out}}$, return the current pair; otherwise, repeat.

Python codes: `lagrange_newton`, `lagrange_newton_step`.

Theory. If $\mathbf{F}(\mathbf{z}^*) = 0$, $\mathbf{J}_F(\mathbf{z}^*)$ is nonsingular, $\mathbf{J}_F$ is locally Lipschitz, and $\mathbf{z}^{(0)}$ is sufficiently close to $\mathbf{z}^*$,

$$\|\mathbf{z}^{(k+1)} - \mathbf{z}^*\| \le C\|\mathbf{z}^{(k)} - \mathbf{z}^*\|^2.$$

A common test problem

$$\min_{\mathbf{x}\in\mathbf{R}^2} f(\mathbf{x}) = \tfrac{1}{2}[(x_1-1)^2 + (x_2-2)^2] \quad \text{s. t.} \quad g(\mathbf{x}) = x_1^2 + x_2 - 1 = 0.$$

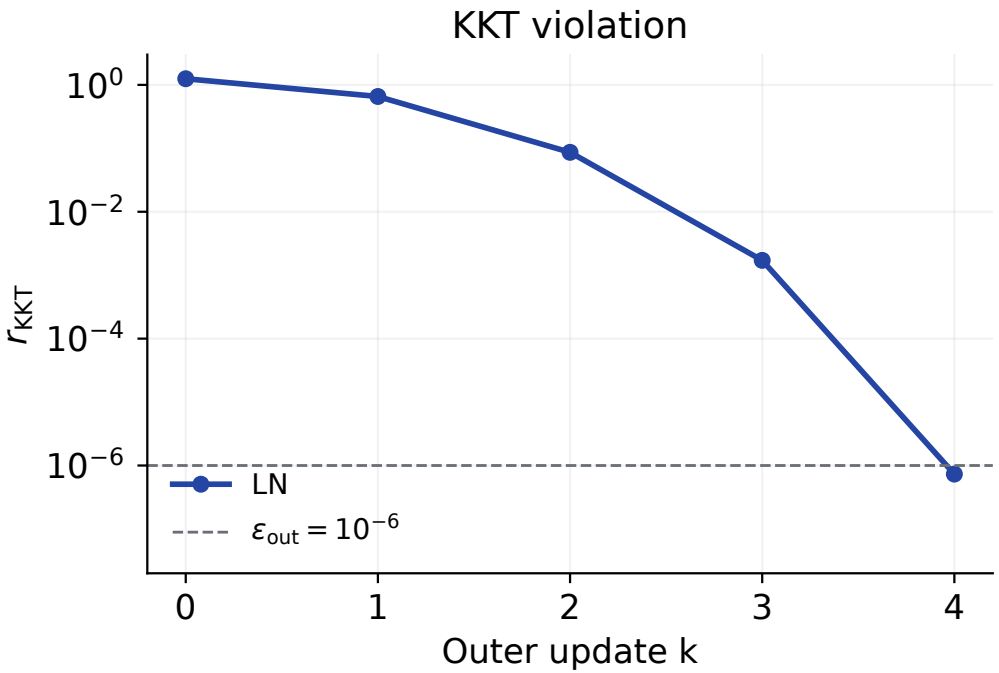
$$\nabla f = \begin{pmatrix} x_1 - 1 \\ x_2 - 2 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 2x_1 \\ 1 \end{pmatrix}, \quad \mathbf{H}_L = \begin{pmatrix} 1+2\lambda & 0 \\ 0 & 1 \end{pmatrix}.$$

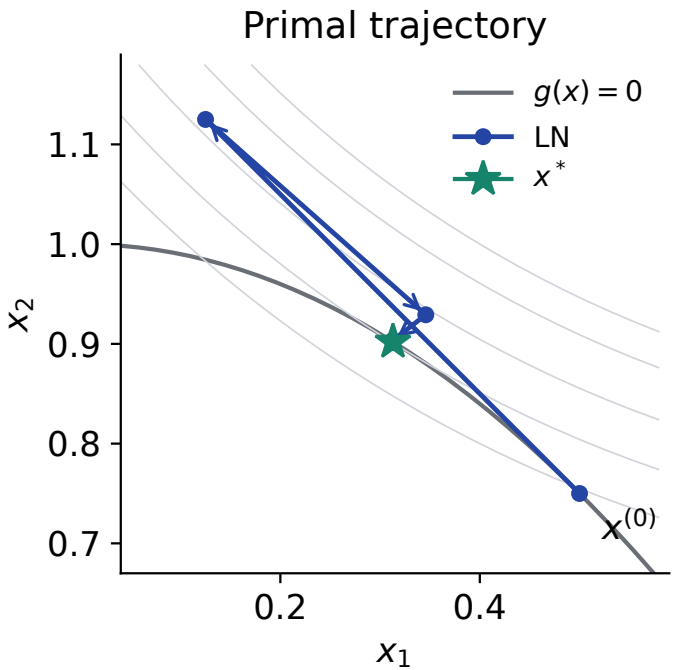
- Start from $\mathbf{x}^{(0)} = (1/2, 3/4)^T$, $\lambda^{(0)} = 0$.
- Eliminating $x_2 = 1 - x_1^2$ gives a strictly convex scalar objective:

$$\frac{df(x_1, 1-x_1^2)}{dx_1} = 2x_1^3 + 3x_1 - 1.$$

$$\boxed{\mathbf{x}^* \approx (0.312908, 0.902088)^T, \qquad \lambda^* \approx 1.097912.}$$

LN on the common test: residuals and trajectory





$\varepsilon_{\mathrm{out}} = 10^{-6}$: four updates / four KKT solves.
Final $r_{\mathrm{stat}} = 7.322 \times 10^{-7}$, $r_{\mathrm{feas}} = 8.160 \times 10^{-8}$.

Limitation 1: the Newton matrix can be singular

Let $\mathbf{A}$ have full column rank (LICQ), and let the columns of $\mathbf{Z}$ form a basis of $\ker(\mathbf{A}^T)$. Then

$$\begin{pmatrix} \mathbf{H}_L & \mathbf{A} \\ \mathbf{A}^T & 0 \end{pmatrix} \text{ nonsingular} \iff \mathbf{Z}^T\mathbf{H}_L\mathbf{Z} \text{ nonsingular.}$$

- Sufficient conditions at a KKT point:

$$\text{KKT SOSC } (\mathbf{Z}^T\mathbf{H}_L\mathbf{Z} \succ 0), \qquad \mathbf{H}_L \succ 0.$$

- Sufficient and necessary conditions at a local minimum:

$$\text{nonsingularity} \iff \text{KKT SOSC.}$$

Limitation 1: the Newton matrix can be singular

Let $\mathbf{A}$ have full column rank (LICQ), and let the columns of $\mathbf{Z}$ form a basis of $\ker(\mathbf{A}^T)$. Then

$$\begin{pmatrix} \mathbf{H}_L & \mathbf{A} \\ \mathbf{A}^T & 0 \end{pmatrix} \text{ nonsingular} \iff \mathbf{Z}^T\mathbf{H}_L\mathbf{Z} \text{ nonsingular.}$$

- Sufficient conditions at a KKT point:

$$\text{KKT SOSC } (\mathbf{Z}^T\mathbf{H}_L\mathbf{Z} \succ 0), \qquad \mathbf{H}_L \succ 0.$$

- Sufficient and necessary conditions at a local minimum:

$$\text{nonsingularity} \iff \text{KKT SOSC.}$$

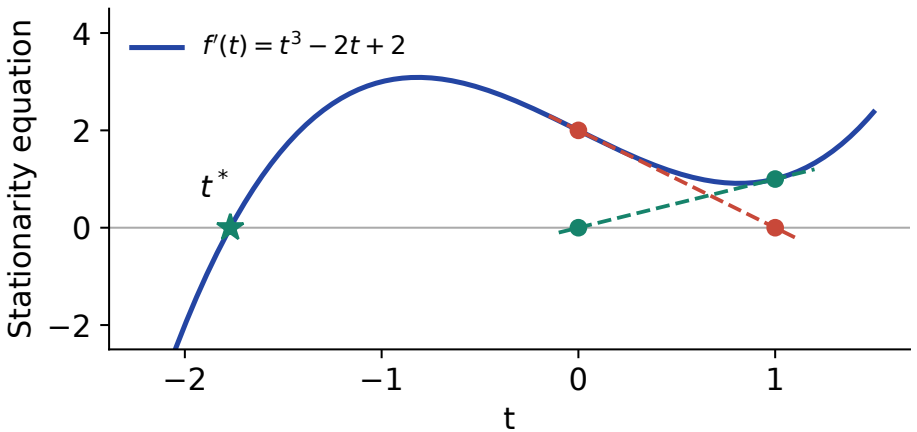
At an arbitrary KKT point, nonsingularity alone says nothing about the sign of curvature (back to this point soon and in the dual part).

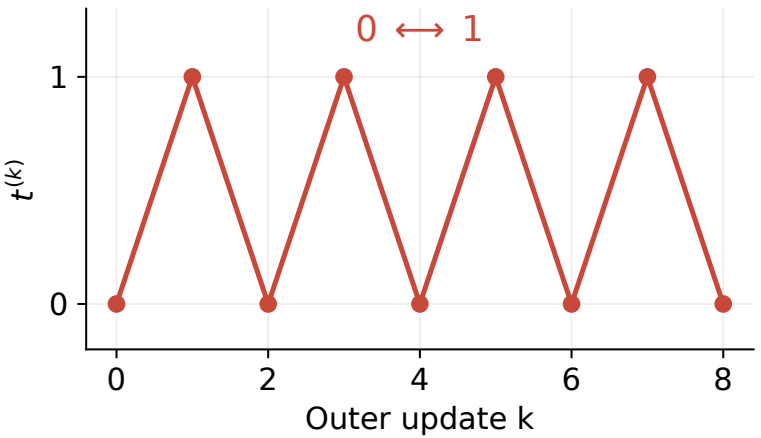
Limitation 2: a distant initial point can cycle

$$\min_{(t,u)} \ \tfrac{1}{4}t^4 - t^2 + 2t + \tfrac{1}{2}u^2 \quad \text{s. t.} \quad u = 0.$$

Starting at $u = \lambda = 0$, LN reduces to

$$t^+ = t - \frac{t^3 - 2t + 2}{3t^2 - 2}, \qquad \boxed{0 \longmapsto 1 \longmapsto 0 \longmapsto \cdots .}$$





Both Newton matrices on the cycle are nonsingular; the minimizer is $t^* \approx -1.7693$.

Limitation 3: why not line-search on the Lagrangian?

For $\mathbf{z} = (\mathbf{x}, \boldsymbol{\lambda})$,

$$\nabla_{\mathbf{z}} L = \begin{pmatrix} \nabla_{\mathbf{x}} L \\ \mathbf{g} \end{pmatrix} = \mathbf{F}, \qquad \nabla^2_{\mathbf{zz}} L = \begin{pmatrix} \mathbf{H}_L & \mathbf{A} \\ \mathbf{A}^T & 0 \end{pmatrix}.$$

Thus LN is Newton's method for $\nabla_{\mathbf{z}} L = \mathbf{0}$.

Limitation 3: why not line-search on the Lagrangian?

For $\mathbf{z} = (\mathbf{x}, \boldsymbol{\lambda})$,

$$\nabla_{\mathbf{z}} L = \begin{pmatrix} \nabla_{\mathbf{x}} L \\ \mathbf{g} \end{pmatrix} = \mathbf{F}, \qquad \nabla^2_{\mathbf{zz}} L = \begin{pmatrix} \mathbf{H}_L & \mathbf{A} \\ \mathbf{A}^T & 0 \end{pmatrix}.$$

Thus LN is Newton's method for $\nabla_{\mathbf{z}} L = \mathbf{0}$.

- A constrained minimum need not minimize L jointly in $(\mathbf{x}, \boldsymbol{\lambda})$.
- The zero multiplier block and primal–multiplier coupling produce indefinite curvature when equality gradients are nonzero.

$$\min_x \ \frac{1}{2} x^2, \quad \text{s. t.} \quad x = 0.$$

$$\mathbf{H}_L = 1, \qquad \mathbf{A} = 1, \qquad \nabla^2_{\mathbf{zz}} L = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

This saddle structure will reappear in the duality lectures (later).

From LN to SQP: a variable change

$$\begin{pmatrix} \mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d}^{(k)} \\ \boldsymbol{\delta}^{(k)} \end{pmatrix} = -\begin{pmatrix} \nabla_{\mathbf{x}} L^{(k)} \\ \mathbf{g}^{(k)} \end{pmatrix}.$$

The first LN equation is

$$\mathbf{H}_L^{(k)} \mathbf{d}^{(k)} + \mathbf{A}^{(k)} \boldsymbol{\delta}^{(k)} = -\nabla f^{(k)} - \mathbf{A}^{(k)} \boldsymbol{\lambda}^{(k)}.$$

From LN to SQP: a variable change

$$\begin{pmatrix} \mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d}^{(k)} \\ \boldsymbol{\delta}^{(k)} \end{pmatrix} = -\begin{pmatrix} \nabla_{\mathbf{x}} L^{(k)} \\ \mathbf{g}^{(k)} \end{pmatrix}.$$

The first LN equation is

$$\mathbf{H}_L^{(k)} \mathbf{d}^{(k)} + \mathbf{A}^{(k)} \boldsymbol{\delta}^{(k)} = -\nabla f^{(k)} - \mathbf{A}^{(k)} \boldsymbol{\lambda}^{(k)}.$$

Set $\widehat{\boldsymbol{\lambda}}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \boldsymbol{\delta}^{(k)}$. Then

$$\boxed{\begin{pmatrix} \mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d}^{(k)} \\ \widehat{\boldsymbol{\lambda}}^{(k+1)} \end{pmatrix} = -\begin{pmatrix} \nabla f^{(k)} \\ \mathbf{g}^{(k)} \end{pmatrix}.}$$

- The primal direction is unchanged.
- We now solve directly for a new multiplier estimate.

From LN to SQP: a variable change (cont'd)

The transformed equations are the KKT conditions of

$$\boxed{\begin{array}{l}\min\limits_{\mathbf{d}} \\ \text{s. t.}\end{array}} \Longrightarrow \begin{pmatrix} \mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d}^{(k)} \\ \widehat{\boldsymbol{\lambda}}^{(k+1)} \end{pmatrix} = -\begin{pmatrix} \nabla f^{(k)} \\ \mathbf{g}^{(k)} \end{pmatrix}.$$

From LN to SQP: a variable change (cont'd)

The transformed equations are the KKT conditions of

$$\boxed{\begin{array}{ll}\min\limits_{\mathbf{d}} & \\ \text{s. t.} & \mathbf{g}^{(k)} + (\mathbf{A}^{(k)})^T\mathbf{d} = 0.\end{array}} \Longrightarrow \begin{pmatrix} \mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d}^{(k)} \\ \widehat{\boldsymbol{\lambda}}^{(k+1)} \end{pmatrix} = -\begin{pmatrix} \nabla f^{(k)} \\ \mathbf{g}^{(k)} \end{pmatrix}.$$

From LN to SQP: a variable change (cont'd)

The transformed equations are the KKT conditions of

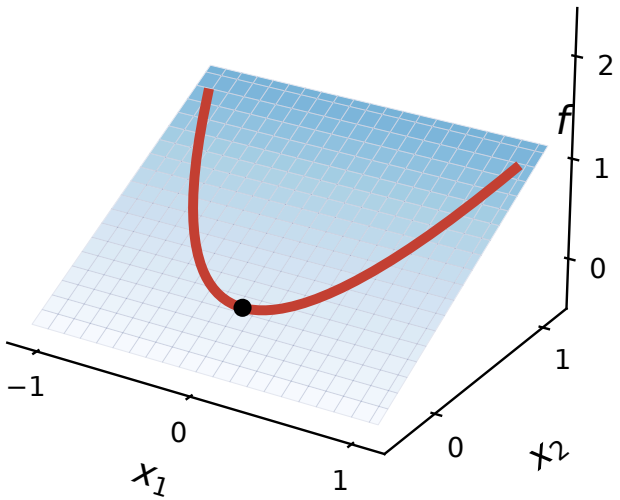
$$\boxed{\begin{array}{ll}\min\limits_{\mathbf{d}} & (\nabla f^{(k)})^T\mathbf{d} + \frac{1}{2}\mathbf{d}^T\mathbf{H}_L^{(k)}\mathbf{d} \\ \text{s. t.} & \mathbf{g}^{(k)} + (\mathbf{A}^{(k)})^T\mathbf{d} = 0.\end{array}} \Longrightarrow \begin{pmatrix}\mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0\end{pmatrix}\begin{pmatrix}\mathbf{d}^{(k)} \\ \widehat{\boldsymbol{\lambda}}^{(k+1)}\end{pmatrix} = -\begin{pmatrix}\nabla f^{(k)} \\ \mathbf{g}^{(k)}\end{pmatrix}.$$

- Quadratic objective + linear constraints = a quadratic program.
- Sequential quadratic programming (SQP):
  - quadratic programs as subproblems;
  - **modifications from the optimization viewpoint**.

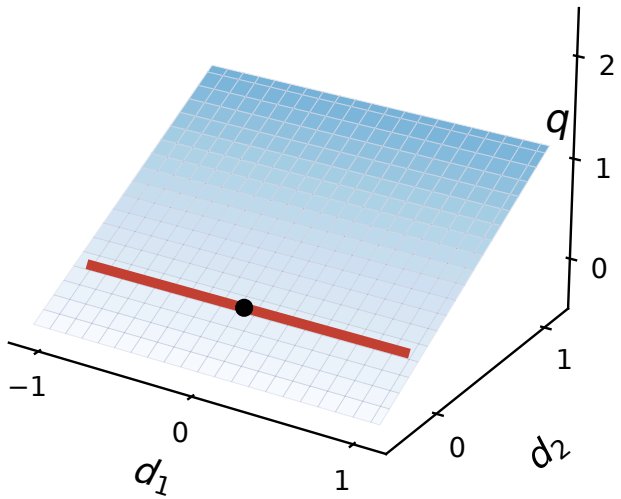
Linearization removes constraint curvature

$$\min_{\mathbf{x}} f(\mathbf{x}) = x_2 \quad \text{s. t.} \quad x_1^2 - x_2 = 0, \qquad \mathbf{x}^* = 0, \quad \lambda^* = 1.$$

Original: curved constraint



Using $H_f = 0$: flat model

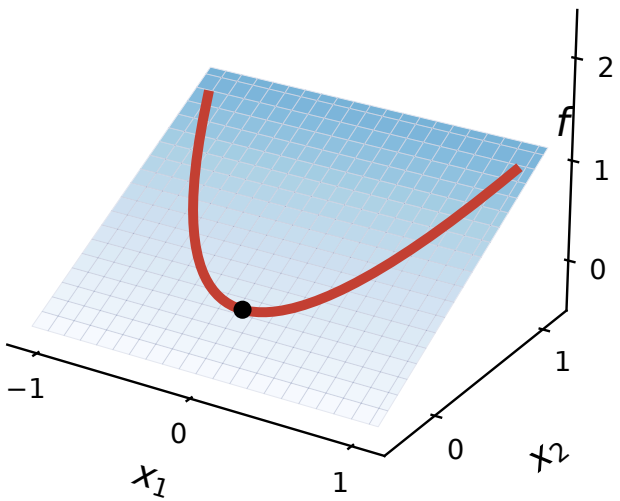


- Original red curve: $(x_1, x_2) = (t, t^2)$, so its objective height is t^2.
- Linearize at zero and use only $\mathbf{H}_f = 0$: $q_f(\mathbf{d}) = d_2$, $d_2 = 0$. The red line has constant height zero: the quadratic model loses the minimum's curvature.

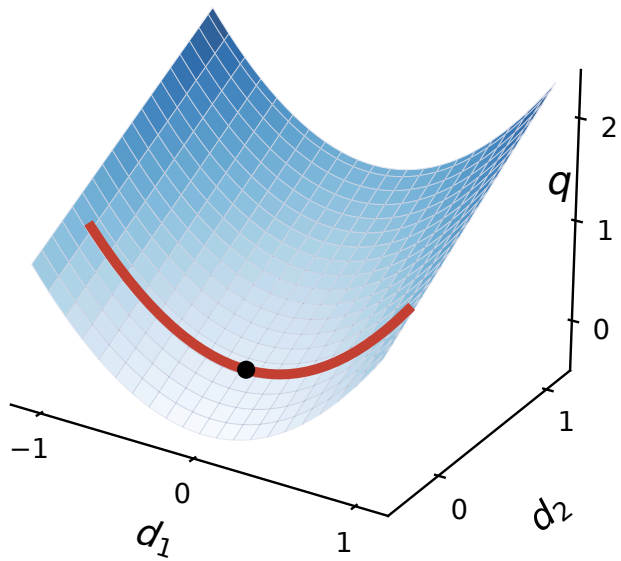
The Lagrangian Hessian restores the missing curvature

$$\mathbf{H}_L = \mathbf{H}_f + \lambda^* \mathbf{H}_g = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \min_{\mathbf{d}} \; q_L(\mathbf{d}) = d_2 + d_1^2 \quad \text{s. t.} \quad d_2 = 0.$$

Original: curved constraint



Using $\mathbf{H}_L$: curvature restored



- The original objective is a plane; the SQP objective is curved.
- Along their red feasible curves, both objective heights are t^2. The curved constraint has become a straight constraint plus objective curvature.

When does the SQP subproblem have a unique minimizer?

Conclusion follows from the analysis for LN.

$$\boxed{\begin{array}{ll}\min\limits_{\mathbf{d}} & (\nabla f^{(k)})^T\mathbf{d} + \frac{1}{2}\mathbf{d}^T\mathbf{H}_L^{(k)}\mathbf{d} \\ \text{s. t.} & \mathbf{g}^{(k)} + (\mathbf{A}^{(k)})^T\mathbf{d} = 0.\end{array}} \Longrightarrow \begin{pmatrix}\mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0\end{pmatrix}\begin{pmatrix}\mathbf{d}^{(k)} \\ \widehat{\boldsymbol{\lambda}}^{(k+1)}\end{pmatrix} = -\begin{pmatrix}\nabla f^{(k)} \\ \mathbf{g}^{(k)}\end{pmatrix}.$$

Suppose LICQ holds at $\mathbf{x}^{(k)}$, and $\ker((\mathbf{A}^{(k)})^T) = \text{Range}(\mathbf{Z}^{(k)})$.

- Sufficient conditions: $(\mathbf{Z}^{(k)})^T\mathbf{H}_L^{(k)}\mathbf{Z}^{(k)} \succ 0$, or even $\mathbf{H}_L^{(k)} \succ 0$.

When does the SQP subproblem have a unique minimizer?

Conclusion follows from the analysis for LN.

$$\boxed{\begin{array}{ll}\min\limits_{\mathbf{d}} & (\nabla f^{(k)})^T\mathbf{d} + \frac{1}{2}\mathbf{d}^T\mathbf{H}_L^{(k)}\mathbf{d} \\ \text{s. t.} & \mathbf{g}^{(k)} + (\mathbf{A}^{(k)})^T\mathbf{d} = 0.\end{array}} \Longrightarrow \begin{pmatrix}\mathbf{H}_L^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0\end{pmatrix}\begin{pmatrix}\mathbf{d}^{(k)} \\ \widehat{\boldsymbol{\lambda}}^{(k+1)}\end{pmatrix} = -\begin{pmatrix}\nabla f^{(k)} \\ \mathbf{g}^{(k)}\end{pmatrix}.$$

Suppose LICQ holds at $\mathbf{x}^{(k)}$, and $\ker((\mathbf{A}^{(k)})^T) = \text{Range}(\mathbf{Z}^{(k)})$.

- Sufficient conditions: $(\mathbf{Z}^{(k)})^T\mathbf{H}_L^{(k)}\mathbf{Z}^{(k)} \succ 0$, or even $\mathbf{H}_L^{(k)} \succ 0$.

Warning. (1) An algebraic KKT solution need not minimize an indefinite model. (2) The SQP subproblem can be infeasible.

SQP supplies two tools beyond LN

- **Control curvature in the primal model:**

$$\mathbf{B}^{(k)} = \mathbf{H}_L^{(k)} + \tau \mathbf{I} \quad \text{or an appropriate Hessian approximation.}$$

A sufficiently large shift makes $(\mathbf{Z}^{(k)})^T \mathbf{B}^{(k)} \mathbf{Z}^{(k)} \succ 0$.

SQP supplies two tools beyond LN

- **Control curvature in the primal model:**

$$\mathbf{B}^{(k)} = \mathbf{H}_L^{(k)} + \tau \mathbf{I} \quad \text{or an appropriate Hessian approximation.}$$

A sufficiently large shift makes $(\mathbf{Z}^{(k)})^T \mathbf{B}^{(k)} \mathbf{Z}^{(k)} \succ 0$.

- **Control step acceptance with an ℓ_1 merit function:**

$$\boxed{\phi_\rho(\mathbf{x}) = f(\mathbf{x}) + \rho \|\mathbf{g}(\mathbf{x})\|_1.}$$

This balances objective improvement against constraint violation.

SQP supplies two tools beyond LN

- **Control curvature in the primal model:**

$$\mathbf{B}^{(k)} = \mathbf{H}_L^{(k)} + \tau \mathbf{I} \quad \text{or an appropriate Hessian approximation.}$$

A sufficiently large shift makes $(\mathbf{Z}^{(k)})^T \mathbf{B}^{(k)} \mathbf{Z}^{(k)} \succ 0$.

- **Control step acceptance with an ℓ_1 merit function:**

$$\boxed{\phi_\rho(\mathbf{x}) = f(\mathbf{x}) + \rho \|\mathbf{g}(\mathbf{x})\|_1.}$$

This balances objective improvement against constraint violation.

Theory I: a feasible local minimum of ϕ_ρ satisfies KKT for some multiplier $\boldsymbol{\lambda}^*$. Conversely, KKT + SOSC and $\rho > \|\boldsymbol{\lambda}^*\|_\infty$ imply a strict local minimum of ϕ_ρ.

SQP supplies two tools beyond LN

- **Control curvature in the primal model:**

$$\mathbf{B}^{(k)} = \mathbf{H}_L^{(k)} + \tau \mathbf{I} \quad \text{or an appropriate Hessian approximation.}$$

A sufficiently large shift makes $(\mathbf{Z}^{(k)})^T \mathbf{B}^{(k)} \mathbf{Z}^{(k)} \succ 0$.

- **Control step acceptance with an ℓ_1 merit function:**

$$\boxed{\phi_\rho(\mathbf{x}) = f(\mathbf{x}) + \rho \|\mathbf{g}(\mathbf{x})\|_1.}$$

This balances objective improvement against constraint violation.

Theory I: a feasible local minimum of ϕ_ρ satisfies KKT for some multiplier $\boldsymbol{\lambda}^*$. Conversely, KKT + SOSC and $\rho > \|\boldsymbol{\lambda}^*\|_\infty$ imply a strict local minimum of ϕ_ρ.

Theory II: after solving the model, $\mathbf{B}^{(k)} \succ 0$, $\mathbf{d}^{(k)} \neq \mathbf{0}$, $\rho > \|\widehat{\boldsymbol{\lambda}}^{(k+1)}\|_\infty$ $\Rightarrow$ merit descent $\Rightarrow$ Armijo accepts small steps.

(Damped) SQP: pseudocode

Choose $(\mathbf{x}^{(0)}, \boldsymbol{\lambda}^{(0)})$, ε_{out}, fixed merit parameter $\rho > 0$, and $\sigma, \beta \in (0,1)$.

1. Choose $\mathbf{B}^{(k)}$; check the linearized constraints and model curvature.
2. Solve for the step and *new* multiplier:

$$\boxed{\begin{pmatrix} \mathbf{B}^{(k)} & \mathbf{A}^{(k)} \\ (\mathbf{A}^{(k)})^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d}^{(k)} \\ \widehat{\boldsymbol{\lambda}}^{(k+1)} \end{pmatrix} = -\begin{pmatrix} \nabla f^{(k)} \\ \mathbf{g}^{(k)} \end{pmatrix}.}$$

3. If $(\mathbf{x}^{(k)}, \widehat{\boldsymbol{\lambda}}^{(k+1)})$ already passes the KKT test, return it. Otherwise set $\boldsymbol{\delta}^{(k)} = \widehat{\boldsymbol{\lambda}}^{(k+1)} - \boldsymbol{\lambda}^{(k)}$, and compute $D^{(k)} = (\nabla f^{(k)})^T \mathbf{d}^{(k)} - \rho \|\mathbf{g}^{(k)}\|_1$.

$$\phi_\rho(\mathbf{x} + \alpha \mathbf{d}) - \phi_\rho(\mathbf{x}) = \alpha D + O(\alpha^2 \|\mathbf{d}\|^2), \quad D = \nabla f^T \mathbf{d} - \rho \|\mathbf{g}\|_1.$$

(Damped) SQP: pseudocode (cont'd)

4. If $D^{(k)} \geq 0$, report failure and stop. Otherwise backtrack from $\alpha = 1$ until the trial point is valid and

$$\boxed{\phi_\rho(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) \leq \phi_\rho(\mathbf{x}^{(k)}) + \sigma \alpha D^{(k)}.}$$

5. Update both components:

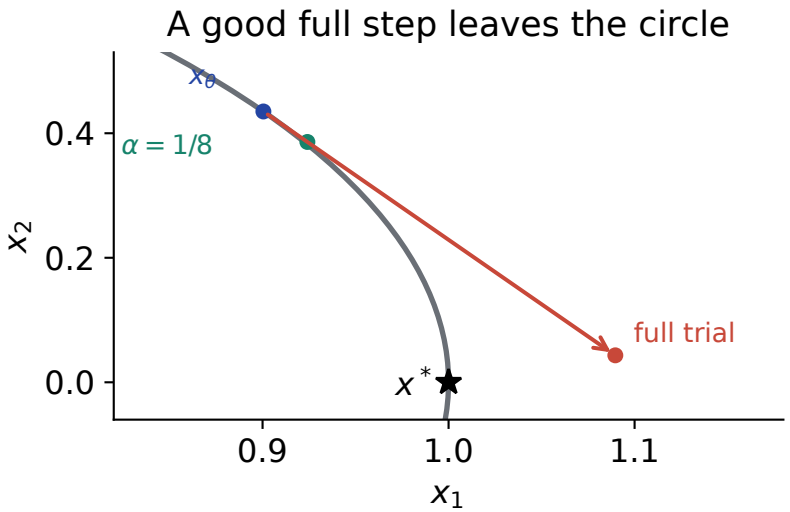
$$\boxed{\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}, \quad \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha \boldsymbol{\delta}^{(k)}.}$$

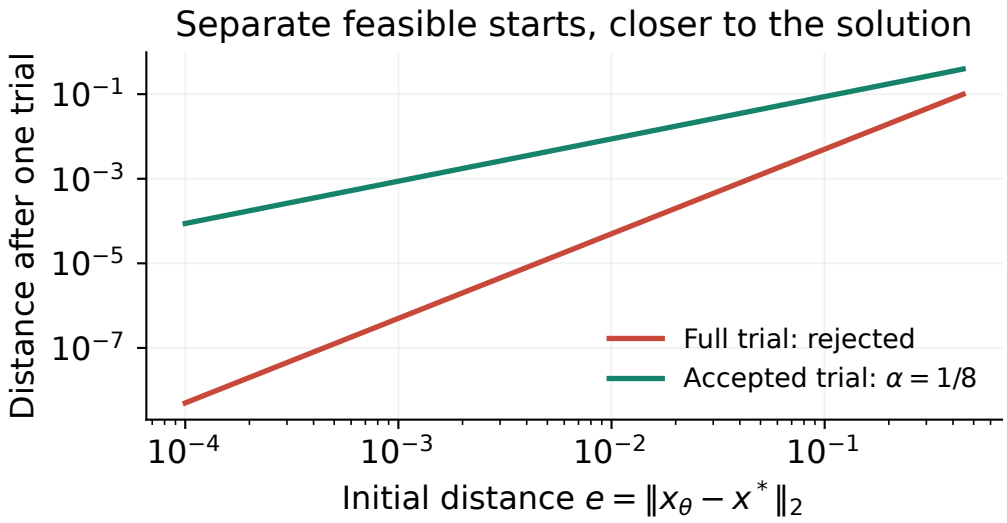
6. Repeat; check the final returned pair too.

Python codes: `damped_sqp`.

A remaining limitation: the Maratos effect

$$f = 2(\|\mathbf{x}\|^2 - 1) - x_1, \quad g = \|\mathbf{x}\|^2 - 1 = 0, \quad \mathbf{x}^* = (1, 0)^T.$$





For $0 < |\theta| \ll 1$, at feasible $\mathbf{x}_\theta = (\cos\theta, \sin\theta)^T$, $\lambda = -3/2$, the exact step satisfies

$$\|\mathbf{x}_\theta + \mathbf{d} - \mathbf{x}^*\| = \tfrac{1}{2}\|\mathbf{x}_\theta - \mathbf{x}^*\|^2, \quad \phi_\rho(\mathbf{x}_\theta + \mathbf{d}) - \phi_\rho(\mathbf{x}_\theta) = (1 + \rho)\sin^2\theta > 0.$$

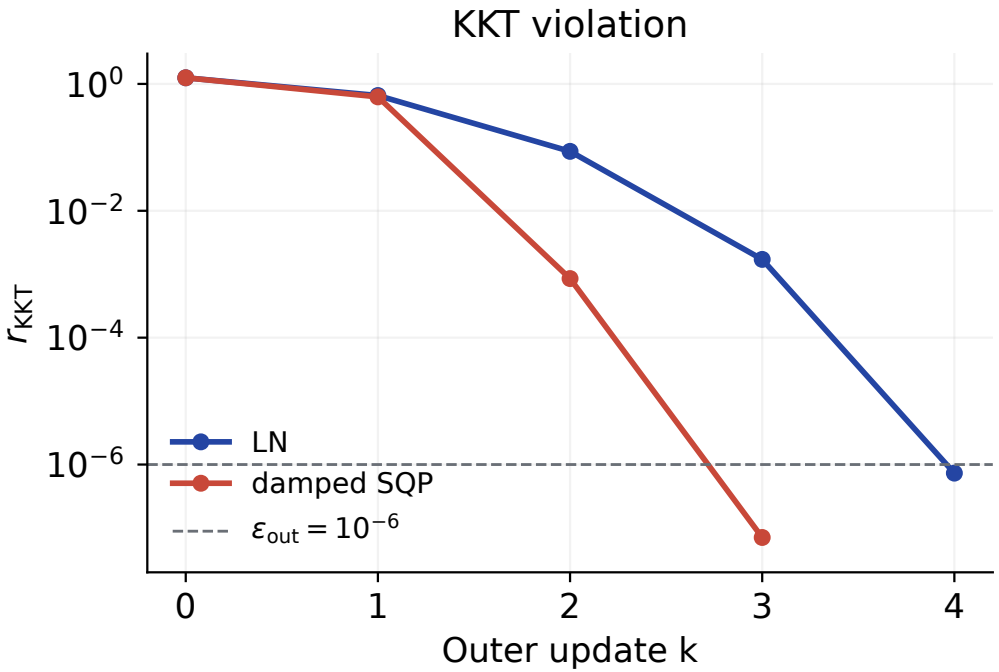
Arbitrarily close to the solution, a quadratically good full step can still be rejected. The graph uses independent starts with $\rho = 2$, $\sigma = 10^{-4}$, and $\beta = 1/2$.

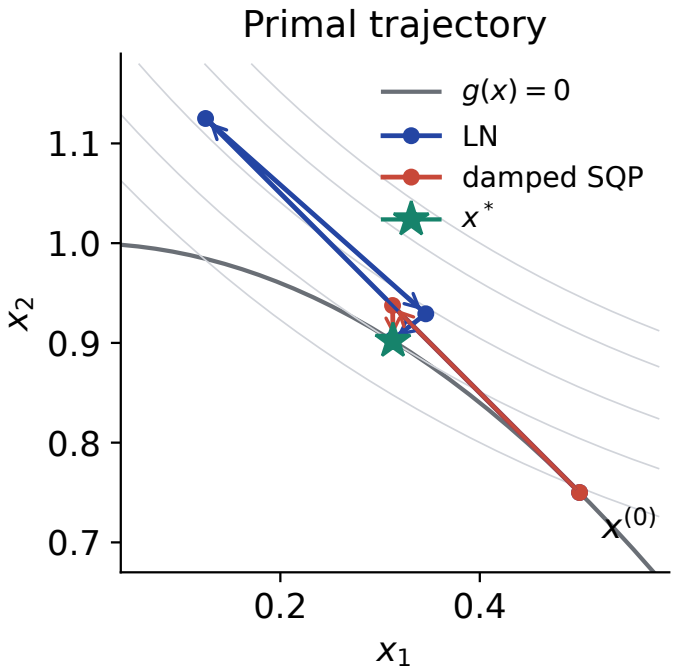
Compare LN and SQP on the same test

$$f = \tfrac{1}{2}[(x_1 - 1)^2 + (x_2 - 2)^2], \qquad g = x_1^2 + x_2 - 1.$$

- Same initial pair: $\mathbf{x}^{(0)} = (1/2, 3/4)^T$, $\lambda^{(0)} = 0$.
- Same outer stopping rule: $r_{\mathrm{KKT}} \le \varepsilon_{\mathrm{out}} = 10^{-6}$.
- Direct KKT solves; normalized linear residual at most $\varepsilon_{\mathrm{in}}^{(k)} \equiv 10^{-10}$.
- LN: exact $\mathbf{H}_L$, full steps.
- SQP: exact $\mathbf{H}_L$ ($\tau = 0$); fixed merit $\rho = 2$, $\sigma = 10^{-4}$, and $\beta = 1/2$.

Common test: KKT violation and primal trajectories





Method	KKT solves	Step lengths	r_{stat}	r_{feas}
LN	4	$1, 1, 1, 1$	$7.322\,10^{-7}$	$8.160\,10^{-8}$
SQP	3	$1/2, 1, 1$	$7.082\,10^{-8}$	$5.670\,10^{-8}$

Comparing QP/ALM and LN/SQP

	QP / ALM	LN / SQP
Target	Original KKT violation	
Accuracy	Balance $\varepsilon_{\text{in}}^{(k)}$ and ε_{out}	
Inner task	Unconstrained minimization	Linear solve / quadratic program
Parameters	Penalty affects inner conditioning	SQP merit penalty and model regularization
Checks	Inner minimization and warm starts	Rank, reduced curvature, indefiniteness, and acceptance

Extension: inequality constraints

Let $\mathbf{g} = 0$, $\mathbf{h} \leq 0$, and $\mathbf{C} = [\nabla h_1, \ldots, \nabla h_p]$.

$$\begin{aligned} \min_{\mathbf{d}} \quad & \nabla f^T \mathbf{d} + \tfrac{1}{2} \mathbf{d}^T \mathbf{B} \mathbf{d} \\ \text{s. t.} \quad & \mathbf{g} + \mathbf{A}^T \mathbf{d} = 0, \qquad \mathbf{h} + \mathbf{C}^T \mathbf{d} \leq 0. \end{aligned}$$

Its KKT conditions include

$$\nabla f + \mathbf{B}\mathbf{d} + \mathbf{A}\widehat{\boldsymbol{\lambda}} + \mathbf{C}\widehat{\boldsymbol{\nu}} = 0, \quad \widehat{\boldsymbol{\nu}} \geq 0, \quad \widehat{\nu}_j (h_j + \nabla h_j^T \mathbf{d}) = 0.$$

Complementarity adds significant complications.

SQP may need adaptive strategies to "guess" active sets.

Take-home messages

- (Still mainly on) equality-constrained optimization.
- LN: KKT equations + Newton linearization.
- SQP: quadratic programs as subproblems + toppings:
  - objective: curvatures from constraints ($\mathbf{H}_L$);
  - constraints: linearized constraints;
  - rank, well-posedness, regularization;
  - step acceptance: merit line search and the Maratos effect.

Much more complex... but one of the most popular methods!